Ordinary Differential Equations

— *EYNTKA* Series —

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Contents

Prerequisites

It is assumed that the reader is comfortable with the concepts introduced in a first course in analysis and linear algebra, importantly that smooth functions and vector-spaces are comfortable concepts.

Introduction Differential Equations

To avoid super-script counting, the book shall work with smooth functions (or else the order of differentiability must be kept track). This shall not change the theory for ODE's and PDEs, especially since later it shall be shown that even working with non-smooth differentiable functions shall end up having the same theory (due to the advent of Distribution Theory [1])

Let $f : \mathbb{R} \to \mathbb{R}$. When working smooth functions, we may take the derivative of the function. This gives the following function taking smooth functions and returning smooth functions:

$$f \mapsto f'$$

Due to the fundamental theorem of calculus, the inverse of this procedure is called *integration*, and gives a unique inverse up to a constant:

$$f \mapsto F + c \qquad c \in \mathbb{R}$$

where F may be defined as $F(x) = \int_t^x f(x)dx$ for appropriate t. Using this result, it is possible to find the smooth function that satisfies an arbitrary degree of differentiation, for example:

$$f^{(n)} = 0$$

which is a polynomial of degree n, or more generally

$$f^{(n)} = g \qquad \text{equiv.} \qquad f^{(n)} - g = 0$$

for some smooth functions g[2].

Given this simple scenario, the question may arise if we may solve more complicated algebraic expression. For example, does there exist a solution to any of these:

$$f = f' \qquad f' + f = 0 \qquad f'' + f' + f = 0$$

[1] see chapter ref:HERE

[2] Note that the solution may not always come in an elementary form, that is expressible as a polynomial-combination and composition of x, log, e^x, or a trigonometric identity

More precisely, we ask if any arbitrary algebraic expression on a smooth function f combined by using the 4 usual arithmetic operators $(+, -, \times, \text{div})$ along with differentiation yields a solution:

$$p(f) = 0$$

more commonly written as:

$$F(x, f(x), f'(x), ..., f^{(n)}(x)) = 0$$

The study of the solution to such equations is called the study of *Ordinary Differential Equations* (ODEs). If we have a smooth function of the form $f : \mathbb{R} \to \mathbb{R}^m$, this is still the study of ODEs, where it is solving a *system of ODEs*. When working with multi-variable functions and partial derivatives, the study of the solution to such equations is called the study of *Partial Differential Equations* (PDEs). The study of ODEs can be thought of as a special case of the study of PDEs.

This book shall start with the study of ODEs in the first chapter, and proceed with the rest of the chapter with the study of PDEs. Only one chapter is dedicated to ODEs as they are a solved problem where most ODEs have a general solution that can be found algorithmically, while this is provably not the case for most PDEs.

From a more physical perspective, differential equations will give you how certain parameters relate to each, for PDEs it is usually over time. As a simple example, take:

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial x} \qquad \text{or} \qquad \frac{\partial}{\partial t} - \frac{\partial}{\partial x} = 0 \tag{1}$$

What this equation is giving us is a restriction we require a function $f(t, x)$ to obey, that is:

$$\frac{\partial f}{\partial t} - \frac{\partial f}{\partial x} = 0$$

For example $f(x, t) = x + t$ would satisfy this equation. One key skill to learn when solving problems in PDEs is to be able to "read" such an equation. For example, equation (1) shows that the speed at which our "system" is evolving over time is identically propositional to the speed at which our "system" is moving. Sometimes, it is quite subtle what an equation may represent. For example:

$$\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} = 0$$

is very similar to equation (1), however it represents how a *wave* evolves over time, with behavior that is very different from equation (1).

The reason why finding solutions to PDEs is interesting to me is because the derivative can give lot's of local information about a system, and sometimes that's all the information we have. For example, it is much easier to describe the force of gravity as

$$F = ma$$

where $a = dv/dt$, and $v = d(\text{displacement})/dt = dD/dt$. We can use this equation to construct a gravitational field to how any object at any point will have a certain force on them represented by a vector pointing downwards times the mass of the object, but this "global solution" has a lot more information than is needed. More generally, it is often easier to come up with the "localizations", since they are easier to define or control most of the time.

Also, after seeing how heat equations and diffusion equations are described I have a new motivation: I wonder if at some point these techniques of relating "local information" in describing how it's dynamics evolve (in physics, it would be how it changes in time, in more general scenarios it can be a recursive function or a sequence) will at some point lend itself useful in solving more general problems. For example, let's say we have a particularly complicated mathematical object. Let's say we can solve it in some nice special cases, perhaps with some restricting assumptions. Can we then step-by-step evolve the problem (or model an evolution of the problem) to eliminate the restrictions? Can we localize in some way to simplify it, and glue that information together? These feel like modelling something like a PDE.

As a final motivator, every differential form has a "canonical representative" in that there always exists representative ω such that ω vanishes under the Laplacian: $\Delta\omega = 0$ (functions, and more generally forms, that vanishes under the laplacian are called *harmonic*). This is the beginning of *Hodge Theory*, which is central in studying the cohomology groups of smooth manifolds. In the same vein as geometry but from another perspective, PDEs are also used to model *Ricci flow*, which is about the flow of a manifold given certain parameters as it changes across time.

Ordinary Differential Equations

We focus on solving *Ordinary differential equations* with the hope that we may generalize to the non-linear case. For example:

$$5y'' + 1y' - 3y = 0$$

is a linear differential equation. Since it is equal to 0, it is called a *homogeneous* linear ODE. Since trigonometric functions have very simple derivative properties, they are also studied regularly. Though this might seem limiting, these in fact cover a satisfyingly large sample of scenario's which we may encounter in physics. If your motivation is within pure mathematics, then having a standard repertoire to solve linear and trigonometric ODE's allows us to use these as "local tools" to solve non-linear ODE's or ODE's approximated by a trigonometric sequence[3]. Here is small a list of types of equations and techniques used to solve ODE's:

1. First order ODE:
   (a) Separable: if the equation can be written in the form: $\frac{dy(x)}{dx} = \frac{m(x)}{n(y)}$, or equivalently:
   $$\frac{dy(x)}{dx} + \frac{m(x)}{n(y)} = 0$$
   Then we can solve this ODE
   (b) Linear (Using an integral factor): If the equation can be written in the form:
   $$\frac{dy(x)}{dx} + p(x)y(x) = g(x)$$
   then we can solve this ODE

[3]If you know some more advanced analysis, think of functions being equal to a series of trigonometric functions or polynomials

2. Second order ODE:
   (a) If the ODE has constant coefficients, is linear, and homogeneous:
   $$ay'' + by' + cy = 0$$
   then we can solve this ode, with $y(x) = Ae^{rx}$ for some $r \in \mathbb{C}$ and $A \in \mathbb{C}$.
   (b) If the ODE is Equidimensional (Cauchy-Euler), linear, homogeneous:
   $$ax^2y'' + bxy' + cy = 0$$
   then the solution is of the form $y(x) = Ax^p$ for some constant $p \in \mathbb{C}$.

There are many more that I've not listed here, I just wanted to give a list to give an idea of what was done.

1

First-Order Scalar Equations

The introduction posed equations such as $f = f'$ and $f' + f = 0$ and asked whether solutions exist and how to find them. A scalar first-order equation $y' = f(t, y)$ puts both questions in their sharpest form: it prescribes a slope at every point of the plane, so a solution is a curve whose tangent everywhere matches the prescribed direction. That reading makes existence plausible, since a curve can be grown from any starting point by following the prescribed slopes; Picard-Lindelöf turns the plausibility into a theorem, and a Lipschitz condition is the price.

Once solutions are known to exist and to be unique, the remaining question is which equations can be integrated in closed form. Separable, linear, and exact equations are the three classes that can, and the three techniques are one technique: each recognizes the equation as a derivative that has already been taken, and integrates it back.

1.1 The General Initial Value Problem (IVP)

We begin our study with the scalar first-order ordinary differential equation in the normal form:

$$y' = f(t, y) \tag{1.1}$$

where $y(t)$ is the unknown function and $f : U \to \mathbb{R}$ is a function defined on some open domain $U \subseteq \mathbb{R}^2$. Physically, if we view t as time and y as position, f prescribes the velocity of a particle based on its current time and location. Geometrically, f assigns a slope to every point in U, generating a *slope field* (or direction field). A solution $\varphi(t)$ is a curve whose tangent at every point $(t, \varphi(t))$ matches the slope prescribed by f.

Definition 1.1.1: Initial Value Problem (IVP)

An *Initial Value Problem* consists of an ODE combined with a specified value at a point t_0:

$$\begin{cases} y' = f(t, y) \\ y(t_0) = y_0 \end{cases}$$

A solution to the IVP on an interval I containing t_0 is a differentiable function $\varphi : I \to \mathbb{R}$ such that $\varphi'(t) = f(t, \varphi(t))$ for all $t \in I$ and $\varphi(t_0) = y_0$.

Before we attempt to solve these equations, we must address the fundamental questions of analysis: Does a solution exist? Is it unique?

Theorem 1.1.2: Picard-Lindelöf Existence and Uniqueness

Let $D = \left\{(t, y) \in \mathbb{R}^2 \;\middle|\; |t - t_0| \le a, |y - y_0| \le b\right\}$ be a rectangular region centered at the initial data. Suppose that:

1. $f(t, y)$ is continuous on D.
2. f is *Lipschitz continuous* with respect to y on D. That is, there exists a constant $L > 0$ such that:
$$|f(t, y_1) - f(t, y_2)| \le L|y_1 - y_2| \qquad \forall (t, y_1), (t, y_2) \in D$$

Then there exists a unique solution $y(t)$ to the IVP defined on some interval $|t - t_0| \le h$ (where $h \le a$).

***Proof*:**
We recast the initial value problem as a fixed-point problem. Integrating $y' = f(t, y)$ from t_0 to t and using $y(t_0) = y_0$ shows that a continuous y solves the IVP on $|t - t_0| \le h$ if and only if it satisfies the integral equation

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds.$$

Solutions are therefore precisely the fixed points of the *Picard operator* $(Ty)(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds$.
Step 1: a complete space on which T acts. Since f is continuous on the compact rectangle D, it is bounded there, say $|f| \le M$. Choose $h > 0$ with

$$h \le a, \qquad Mh \le b, \qquad Lh < 1,$$

and let

$$X = \{y \in C([t_0 - h,\, t_0 + h]) \mid |y(t) - y_0| \le b \text{ for all } t\},$$

with the sup metric $d(y_1, y_2) = \sup_t |y_1(t) - y_2(t)|$. Being a closed subset of the complete space $C([t_0 - h, t_0 + h])$, the space (X, d) is itself complete.

Step 2: T maps X into itself. For $y \in X$ the function Ty is continuous, and

$$|(Ty)(t) - y_0| = \left| \int_{t_0}^{t} f(s, y(s))\, ds \right| \leq M\,|t - t_0| \leq Mh \leq b,$$

so $Ty \in X$.

Step 3: T is a contraction. For $y_1, y_2 \in X$ the integrand is controlled by the Lipschitz hypothesis, $|f(s, y_1(s)) - f(s, y_2(s))| \leq L\,|y_1(s) - y_2(s)| \leq L\, d(y_1, y_2)$; integrating over the interval from t_0 to t, of length $|t - t_0| \leq h$,

$$|(Ty_1)(t) - (Ty_2)(t)| \leq \left| \int_{t_0}^{t} |f(s, y_1(s)) - f(s, y_2(s))|\, ds \right| \leq L\,|t - t_0|\, d(y_1, y_2) \leq Lh\, d(y_1, y_2).$$

Taking the supremum over t gives $d(Ty_1, Ty_2) \leq (Lh)\, d(y_1, y_2)$, and $Lh < 1$, so T is a contraction.

By the Banach contraction principle (*Analysis on Metric and Function Spaces* [1]), T has a unique fixed point $y \in X$. This y is the unique solution of the integral equation, hence of the IVP; and since $s \mapsto f(s, y(s))$ is continuous, the fundamental theorem of calculus shows y is continuously differentiable with $y'(t) = f(t, y(t))$.

Note that if $\frac{\partial f}{\partial y}$ exists and is continuous, the Lipschitz condition is automatically satisfied locally.

1.2 Separable Equations

The simplest class of first-order equations are those where the dependence on t and y can be decoupled.

Definition 1.2.1: Separable Equation

A first-order ODE is *separable* if it can be written in the form:

$$\frac{dy}{dt} = g(t)h(y)$$

or equivalently, if $h(y) \neq 0$:

$$\frac{1}{h(y)}\, dy = g(t)\, dt \tag{1.2}$$

The solution technique relies on the chain rule. Integrating (1.2) with respect to t:

$$\int \frac{1}{h(y(t))} \frac{dy}{dt}\, dt = \int g(t)\, dt \implies \int \frac{1}{h(y)}\, dy = \int g(t)\, dt$$

This yields an implicit relationship $H(y) = G(t) + C$.

Example 1.2.2: Solving a Separable IVP

Solve the IVP:

$$y' = -2ty^2, \qquad y(0) = 1$$

Solution: We separate variables (assuming $y \neq 0$):

$$\frac{dy}{y^2} = -2t\,dt$$

Integrating both sides:

$$\int y^{-2}\,dy = \int -2t\,dt \implies -\frac{1}{y} = -t^2 + C \implies y(t) = \frac{1}{t^2 - C}$$

Using the initial condition $y(0) = 1$:

$$1 = \frac{1}{0 - C} \implies C = -1$$

Thus, the solution is $y(t) = \frac{1}{t^2+1}$. Note that this solution is defined for all $t \in \mathbb{R}$. Had the initial condition been $y(0) = -1$, we would have found $y(t) = \frac{1}{t^2-1}$, which blows up at $t = \pm 1$, illustrating that solutions to non-linear ODEs may exist only locally.

Remark on Singular Solutions: In the separation step, we divided by $h(y)$. The roots of $h(y) = 0$ represent constant solutions (equilibrium solutions) $y(t) = c$. These must be checked separately to see if they satisfy the initial conditions or if solutions converge to them.

1.3 Linear Equations and Integrating Factors

Linear equations are the only class of ODEs for which a complete, closed-form general solution is always guaranteed (provided the coefficient functions are continuous).

Definition 1.3.1: First-Order Linear ODE

A first-order ODE is *linear* if it can be written as:

$$y' + p(t)y = g(t)$$

If $g(t) \equiv 0$, the equation is *homogeneous*; otherwise, it is *inhomogeneous*.

To solve this, we seek a function $\mu(t)$, called an *integrating factor*, such that multiplying the equation by $\mu(t)$ transforms the left-hand side into a total derivative of a product.

$$\mu(t)y' + \mu(t)p(t)y = \mu(t)g(t)$$

We want the LHS to be $\frac{d}{dt}[\mu(t)y] = \mu(t)y' + \mu'(t)y$. Matching terms, we require:

$$\mu'(t) = \mu(t)p(t)$$

This is a separable equation for μ: $\frac{d\mu}{\mu} = p(t)dt$, yielding $\mu(t) = \exp\left(\int p(t)\,dt\right)$.

Theorem 1.3.2: General Solution of Linear Equations

The general solution to $y' + p(t)y = g(t)$ is given by:

$$y(t) = \frac{1}{\mu(t)} \left[\int \mu(t)g(t)\,dt + C \right]$$

where $\mu(t) = e^{\int p(t)\,dt}$.

***Proof*:**
Multiply by $\mu(t)$. Then $\frac{d}{dt}[\mu(t)y] = \mu(t)g(t)$. Integrating both sides yields $\mu(t)y = \int \mu(t)g(t)dt + C$. Dividing by $\mu(t)$ gives the result.

Example 1.3.3: Linear Equation with Discontinuous Forcing
Consider $y' + 2y = g(t)$ where $y(0) = 0$ and $g(t) = \chi_{[0,1]}$ (the characteristic function of $[0, 1]$).

Solution: The integrating factor is $\mu(t) = e^{\int 2dt} = e^{2t}$.

$$\frac{d}{dt}[e^{2t}y] = e^{2t}g(t)$$

Integrating from 0 to t:

$$e^{2t}y(t) - e^0y(0) = \int_0^t e^{2s}g(s)\,ds$$

If $0 \le t \le 1$:

$$e^{2t}y(t) = \int_0^t e^{2s}(1)\,ds = \frac{1}{2}(e^{2t} - 1) \implies y(t) = \frac{1}{2}(1 - e^{-2t})$$

If $t > 1$:

$$e^{2t}y(t) = \int_0^1 e^{2s}\,ds + \int_1^t 0\,ds = \frac{1}{2}(e^2 - 1) \implies y(t) = \frac{1}{2}(e^2 - 1)e^{-2t}$$

Notice that while $g(t)$ is discontinuous, the solution $y(t)$ is continuous, though $y'(t)$ is discontinuous at $t = 1$. This "smoothing" property is characteristic of ODEs.

1.4 Exact Equations and Differential Forms

Here we take a slightly more sophisticated view. We can rewrite a first-order equation $y' = f(x, y)$ symmetrically using differential forms:

$$M(x, y)\,dx + N(x, y)\,dy = 0$$

We ask: does there exist a scalar function (a potential) $\psi(x, y)$ such that $d\psi = Mdx + Ndy$? If so, the differential equation becomes $d\psi = 0$, implying the general solution is the level set $\psi(x, y) = C$.

Definition 1.4.1: Exact Differential Equation

The equation $M(x,y)dx + N(x,y)dy = 0$ is *exact* in a region R if there exists a function ψ such that:

$$\frac{\partial \psi}{\partial x} = M \qquad \text{and} \qquad \frac{\partial \psi}{\partial y} = N$$

By equality of mixed partials ($\psi_{xy} = \psi_{yx}$), a necessary condition for exactness is:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \tag{1.3}$$

On a simply connected domain (like a rectangle or all of $\mathbb{R}^2$), the Poincaré Lemma states this condition is also sufficient.

Example 1.4.2: Solving an Exact Equation

Solve $(2xy+3)dx + (x^2-1)dy = 0$.

Check Exactness:

$$\partial_y(2xy+3) = 2x \qquad \partial_x(x^2-1) = 2x$$

The equation is exact. We seek ψ such that $\psi_x = 2xy + 3$. Integrating with respect to x:

$$\psi(x,y) = x^2 y + 3x + h(y)$$

Now differentiate with respect to y and match N:

$$\psi_y = x^2 + h'(y) \stackrel{!}{=} x^2 - 1 \implies h'(y) = -1 \implies h(y) = -y$$

Thus $\psi(x,y) = x^2y + 3x - y$. The implicit solution is $x^2y + 3x - y = C$.

Integrating Factors for Non-Exact Equations

If $Mdx + Ndy = 0$ is not exact, we may look for a function $\mu(x,y)$ such that $(\mu M)dx + (\mu N)dy = 0$ is exact. This requires:

$$(\mu M)_y = (\mu N)_x \implies \mu_y M + \mu M_y = \mu_x N + \mu N_x$$

This is a PDE for μ, which is generally harder to solve than the original ODE! However, if we assume μ depends only on x (i.e., $\mu_y = 0$), the condition simplifies to:

$$\frac{d\mu}{dx} N = \mu(M_y - N_x) \implies \frac{1}{\mu}\frac{d\mu}{dx} = \frac{M_y - N_x}{N}$$

If the RHS depends only on x, we can solve for μ. A similar check can be done assuming $\mu = \mu(y)$.

1.5 Bernoulli Equations

We conclude this chapter with a classic non-linear equation that can be linearized. A *Bernoulli equation* has the form:

$$y' + p(t)y = g(t)y^n \qquad n \in \mathbb{R}, n \neq 0, 1 \tag{1.4}$$

The substitution $v = y^{1-n}$ linearizes this equation.

$$v' = (1-n)y^{-n}y'$$

Multiplying (1.4) by $(1-n)y^{-n}$:

$$(1-n)y^{-n}y' + (1-n)p(t)y^{1-n} = (1-n)g(t)$$

$$v' + (1-n)p(t)v = (1-n)g(t)$$

This is now a standard linear equation for $v(t)$, which can be solved using an integrating factor.

2

Second-Order Linear Equations

Raising the order to two is where the subject becomes linear algebra. The solutions of a homogeneous linear equation are closed under addition and scalar multiplication, so they form a vector space, and dimension, independence, and basis all apply to it: the general solution stops being a formula and becomes a statement about a basis. The Wronskian is the determinant that detects independence of solutions, and Abel's identity computes it without knowing the solutions themselves.

For constant coefficients the basis can be written down, since exponentials reduce the equation to its characteristic polynomial and solving the equation reduces to factoring. Variation of parameters then handles an inhomogeneous forcing term by letting the coefficients in that basis expansion vary with t.

2.1 Definitions and Theoretical Framework

We begin by establishing the precise vocabulary used to classify these equations.

Definition 2.1.1: Second-Order Linear ODE

A second-order ordinary differential equation is called *linear* if it can be written in the form:

$$P(t)y'' + Q(t)y' + R(t)y = G(t) \tag{2.1}$$

where P, Q, R, G are continuous functions on an interval I, and $P(t) \neq 0$ on I.
If we divide by $P(t)$, we obtain the *standard form*:

$$y'' + p(t)y' + q(t)y = g(t)$$

Definition 2.1.2: Homogeneous vs. Inhomogeneous

Referring to the standard form $y'' + p(t)y' + q(t)y = g(t)$:

- If $g(t) \equiv 0$ for all t, the equation is called *homogeneous*.
- If $g(t) \not\equiv 0$, the equation is called *inhomogeneous* (or non-homogeneous).

The distinction is crucial: homogeneous equations describe the "natural" behavior of a system, while inhomogeneous equations describe the system's response to an external "forcing" function $g(t)$. The following theorem guarantees that as long as the coefficient functions $p(t)$ and $q(t)$ are continuous, the solution extends across the entire interval of continuity.

Theorem 2.1.3: Existence and Uniqueness for Linear Equations

Consider the Initial Value Problem (IVP):

$$y'' + p(t)y' + q(t)y = g(t), \qquad y(t_0) = y_0, \quad y'(t_0) = y_0'$$

If $p(t)$, $q(t)$, and $g(t)$ are continuous on an open interval I containing t_0, then there exists a unique solution $y(t)$ defined on the entire interval I.

***Proof*:**
This is a special case of the general Existence and Uniqueness theorem for linear systems (which we will cover in Chapter 3). We convert the second-order scalar equation into a first-order system by setting $x_1 = y$ and $x_2 = y'$. This yields $\vec{x}' = A(t)\vec{x} + \vec{g}(t)$. Since the entries of $A(t)$ (which are $p(t)$ and $q(t)$) are continuous on I, the Picard-Lindelöf theorem for systems guarantees a unique global solution on I.

Example 2.1.4: Interval of Validity
Determine the largest interval on which the IVP has a unique solution:

$$(t^2 - 4)y'' + 3ty' + (\cos t)y = e^t, \qquad y(0) = 1, \quad y'(0) = 0$$

Solution: First, write the equation in standard form by dividing by $(t^2 - 4)$:

$$y'' + \frac{3t}{t^2 - 4}y' + \frac{\cos t}{t^2 - 4}y = \frac{e^t}{t^2 - 4}$$

The coefficients are discontinuous at $t = \pm 2$. The initial point is $t_0 = 0$. The largest interval containing 0 where all functions are continuous is $(-2, 2)$. Thus, the solution is guaranteed to exist uniquely on $t \in (-2, 2)$.

2.2 The Structure of Solutions

The set of solutions to a homogeneous linear ODE forms a vector space.

Theorem 2.2.1: Principle of Superposition

If y_1 and y_2 are solutions to the homogeneous equation $L[y] = y'' + p(t)y' + q(t)y = 0$, then any linear combination $c_1y_1 + c_2y_2$ is also a solution.

***Proof*:**
This follows from the linearity of the differential operator L.

$$L[c_1y_1 + c_2y_2] = c_1L[y_1] + c_2L[y_2] = c_1(0) + c_2(0) = 0$$

To describe *all* solutions, we need a "basis" for this solution space.

Definition 2.2.2: Wronskian

Let y_1, y_2 be two differentiable functions. The *Wronskian* of y_1 and y_2 is the determinant:

$$W[y_1, y_2](t) = \det \begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix} = y_1(t)y_2'(t) - y_2(t)y_1'(t)$$

The Wronskian is crucial because it allows us to test if solutions are linearly independent at a specific point, which is sufficient for linear independence on the entire interval.

Theorem 2.2.3: Abel's Identity

For the ODE $y'' + p(t)y' + q(t)y = 0$, the Wronskian $W(t)$ satisfies the first-order differential equation $W' = -p(t)W$. Thus,

$$W(t) = Ce^{-\int p(t)dt}$$

where C is a constant determined by the initial conditions.

***Proof*:**
Compute the derivative of the Wronskian definition:

$$\begin{aligned} W' &= (y_1y_2' - y_2y_1')' \\ &= (y_1'y_2' + y_1y_2'') - (y_2'y_1' + y_2y_1'') \\ &= y_1y_2'' - y_2y_1'' \end{aligned}$$

Since y_1 and y_2 satisfy the ODE $y'' = -p(t)y' - q(t)y$, substitute this into the expression for W':

$$\begin{aligned} W' &= y_1[-p(t)y_2' - q(t)y_2] - y_2[-p(t)y_1' - q(t)y_1] \\ &= -p(t)(y_1y_2' - y_2y_1') - q(t)(y_1y_2 - y_2y_1) \\ &= -p(t)W - q(t)(0) \\ &= -p(t)W \end{aligned}$$

This is a separable first-order linear equation, the solution of which is $W(t) = Ce^{-\int p(t)dt}$.

Theorem 2.2.4: General Solution of Homogeneous Equations

Let y_1, y_2 be two solutions to the homogeneous equation $y'' + p(t)y' + q(t)y = 0$ on an interval I. If there exists a point $t_0 \in I$ such that $W[y_1, y_2](t_0) \neq 0$, then $\{y_1, y_2\}$ form a *Fundamental Set of Solutions.*
The general solution is:

$$y(t) = c_1 y_1(t) + c_2 y_2(t)$$

***Proof*:**
Let $y(t)$ be any solution. We want to find c_1, c_2 such that $y(t) = c_1 y_1(t) + c_2 y_2(t)$. Consider the initial conditions at t_0: $y(t_0) = Y_0$ and $y'(t_0) = Y_0'$. We set up the linear system:

$$\begin{cases} c_1 y_1(t_0) + c_2 y_2(t_0) = Y_0 \\ c_1 y_1'(t_0) + c_2 y_2'(t_0) = Y_0' \end{cases}$$

The determinant of the coefficient matrix is exactly $W[y_1, y_2](t_0)$. Since $W(t_0) \neq 0$, a unique pair (c_1, c_2) exists. By the Uniqueness Theorem, since $c_1 y_1 + c_2 y_2$ satisfies the ODE and the same initial conditions as $y(t)$, they must be the same function.

Example 2.2.5: Checking Linear Independence
Show that $y_1(t) = e^t$ and $y_2(t) = e^{-t}$ form a fundamental set of solutions for $y'' - y = 0$.

Solution: Check if they solve the ODE:

$$y_1'' - y_1 = (e^t) - e^t = 0 \quad \checkmark \qquad y_2'' - y_2 = (e^{-t}) - e^{-t} = 0 \quad \checkmark$$

Now compute the Wronskian:

$$W(t) = \det \begin{pmatrix} e^t & e^{-t} \\ e^t & -e^{-t} \end{pmatrix} = (e^t)(-e^{-t}) - (e^{-t})(e^t) = -1 - 1 = -2$$

Since $W(t) = -2 \neq 0$, the functions are linearly independent.

2.3 Homogeneous Equations with Constant Coefficients

We now turn to solving specific equations. The simplest case is when P, Q, R are constants.

$$ay'' + by' + cy = 0 \qquad a, b, c \in \mathbb{R}, a \neq 0 \tag{2.2}$$

Motivated by the property that exponentials reproduce themselves under differentiation, we guess a solution of the form $y = e^{rt}$. Substituting this into (2.2):

$$a(r^2 e^{rt}) + b(re^{rt}) + c(e^{rt}) = 0 \implies e^{rt}(ar^2 + br + c) = 0$$

Since $e^{rt} \neq 0$, we must solve the *Characteristic Equation*:

$$ar^2 + br + c = 0 \tag{2.3}$$

The roots are $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Theorem 2.3.1: Solutions to Constant Coefficient Equations

Depending on the discriminant $\Delta = b^2 - 4ac$, we have three cases for the general solution $y(t)$:

1. *Distinct Real Roots ($\Delta > 0$):* Two roots $r_1 \neq r_2$.
$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$
2. *Repeated Real Root ($\Delta = 0$):* One root $r = -b/2a$.
$$y(t) = c_1 e^{rt} + c_2 t e^{rt}$$
3. *Complex Conjugate Roots ($\Delta < 0$):* Roots $r = \lambda \pm i\mu$.
$$y(t) = e^{\lambda t}(c_1 \cos(\mu t) + c_2 \sin(\mu t))$$

***Proof*:**
Case 1 follows directly from the characteristic equation yielding distinct r_1, r_2, and the Wronskian $W[e^{r_1 t}, e^{r_2 t}] = (r_2 - r_1)e^{(r_1+r_2)t} \neq 0$.

Case 2 is proven via reduction of order. Let $y_1 = e^{rt}$. Guess $y_2(t) = v(t)e^{rt}$. Then:
$$y_2' = (v' + rv)e^{rt}, \quad y_2'' = (v'' + 2rv' + r^2 v)e^{rt}$$
Substituting into the ODE $ay'' + by' + cy = 0$:
$$a(v'' + 2rv' + r^2 v) + b(v' + rv) + cv = 0$$
Rearranging terms by derivatives of v:
$$av'' + (2ar + b)v' + (ar^2 + br + c)v = 0$$
Since r satisfies the characteristic equation, the coefficient of v is zero. Since $r = -b/2a$ is a repeated root, $2ar + b = 2a(-b/2a) + b = 0$, so the coefficient of v' is also zero. We are left with $av'' = 0 \implies v(t) = c_1 t + c_2$. Choosing the simplest non-constant solution gives $v(t) = t$, so $y_2 = te^{rt}$.

Case 3 uses Euler's formula $e^{(\lambda + i\mu)t} = e^{\lambda t}(\cos \mu t + i \sin \mu t)$. Since the ODE is linear, the real and imaginary parts of a complex solution are also solutions.

Example 2.3.2: Solving Constant Coefficient Problems
Find the general solution for:

1. $y'' - 5y' + 6y = 0$
2. $y'' + 4y' + 4y = 0$
3. $y'' + 4y' + 13y = 0$

Solutions:

1. Char. Eq: $r^2 - 5r + 6 = (r-2)(r-3) = 0$. Roots: $r_1 = 2, r_2 = 3$.
$$y(t) = c_1 e^{2t} + c_2 e^{3t}$$
2. Char. Eq: $r^2 + 4r + 4 = (r+2)^2 = 0$. Root: $r = -2$ (repeated).
$$y(t) = c_1 e^{-2t} + c_2 t e^{-2t}$$
3. Char. Eq: $r^2 + 4r + 13 = 0$. Using quadratic formula:
$$r = \frac{-4 \pm \sqrt{16-52}}{2} = -2 \pm 3i$$
Here $\lambda = -2, \mu = 3$.
$$y(t) = e^{-2t}(c_1 \cos(3t) + c_2 \sin(3t))$$

2.4 Cauchy-Euler Equations

Definition 2.4.1: Cauchy-Euler Equation

An *Equidimensional* or *Cauchy-Euler* equation is of the form:
$$at^2 y'' + bty' + cy = 0 \qquad t > 0$$

We guess solutions of the form $y = t^m$. Substituting:
$$at^2(m(m-1)t^{m-2}) + bt(mt^{m-1}) + c(t^m) = 0$$
This simplifies to the auxiliary equation:
$$am(m-1) + bm + c = 0 \implies am^2 + (b-a)m + c = 0$$

Theorem 2.4.2: Solutions to Cauchy-Euler Equations

Let m_1, m_2 be the roots of $am^2 + (b-a)m + c = 0$.

1. *Distinct Real Roots:* $y(t) = c_1 t^{m_1} + c_2 t^{m_2}$
2. *Repeated Real Root ($m_1 = m_2 = m$):* $y(t) = c_1 t^m + c_2 t^m \ln(t)$
3. *Complex Roots ($m = \lambda \pm i\mu$):*
$$y(t) = t^\lambda \left[c_1 \cos(\mu \ln t) + c_2 \sin(\mu \ln t)\right]$$

***Proof*:**
Substitute $t = e^x$ (so $x = \ln t$). Then use the chain rule:

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt} = \frac{1}{t}\frac{dy}{dx}$$

$$\frac{d^2y}{dt^2} = \frac{d}{dt}\left(\frac{1}{t}\frac{dy}{dx}\right) = -\frac{1}{t^2}\frac{dy}{dx} + \frac{1}{t}\frac{d^2y}{dx^2}\frac{1}{t} = \frac{1}{t^2}\left(\frac{d^2y}{dx^2} - \frac{dy}{dx}\right)$$

Substituting these into the original ODE yields a constant coefficient equation in x:

$$a\left(\frac{d^2y}{dx^2} - \frac{dy}{dx}\right) + b\frac{dy}{dx} + cy = 0 \implies a\frac{d^2y}{dx^2} + (b-a)\frac{dy}{dx} + cy = 0$$

The auxiliary equation for t is exactly the characteristic equation for x. The solutions in terms of x (e.g., $e^{m_1 x}$) become solutions in terms of t (e.g., $(e^x)^{m_1} = t^{m_1}$).

Example 2.4.3: Cauchy-Euler Application
Solve $t^2y'' - 3ty' + 3y = 0$ for $t > 0$.

Solution: Auxiliary equation: $1m(m-1) - 3m + 3 = 0 \implies m^2 - 4m + 3 = 0$. Factors to $(m-1)(m-3) = 0$. Roots are $1, 3$.

$$y(t) = c_1 t + c_2 t^3$$

2.5 Inhomogeneous Equations

To solve $y'' + p(t)y' + q(t)y = g(t)$, we rely on the following structural theorem.

Theorem 2.5.1: General Solution of Inhomogeneous Equations

The general solution to the inhomogeneous equation is:

$$y(t) = y_h(t) + y_p(t)$$

where $y_h(t)$ is the general homogeneous solution and $y_p(t)$ is *any* particular solution.

***Proof*:**
Let $L[y] = y'' + p(t)y' + q(t)y$. Let Y be any solution such that $L[Y] = g$. Let y_p be a particular solution such that $L[y_p] = g$. Then $L[Y - y_p] = L[Y] - L[y_p] = g - g = 0$. Thus $Y - y_p$ solves the homogeneous equation, so $Y - y_p = y_h$. Hence $Y = y_h + y_p$.

Definition 2.5.2: Variation of Parameters

Given a fundamental set $\{y_1, y_2\}$ for the homogeneous equation, we seek a particular solution of the form:

$$y_p(t) = u_1(t)y_1(t) + u_2(t)y_2(t)$$

Theorem 2.5.3: Variation of Parameters Formula

The particular solution is given by:

$$y_p(t) = -y_1(t) \int \frac{y_2(t)g(t)}{W(t)}\,dt + y_2(t) \int \frac{y_1(t)g(t)}{W(t)}\,dt$$

where $W(t) = W[y_1, y_2](t)$.

***Proof*:**
We differentiate our guess $y_p = u_1 y_1 + u_2 y_2$:

$$y_p' = u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2'$$

To simplify, we impose the condition: $u_1' y_1 + u_2' y_2 = 0$. Then $y_p' = u_1 y_1' + u_2 y_2'$. Differentiating again:

$$y_p'' = u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2''$$

Substitute y_p, y_p', y_p'' into the ODE. Since y_1, y_2 are homogeneous solutions, terms involving $u_1(\dots)$ and $u_2(\dots)$ vanish. We are left with:

$$u_1' y_1' + u_2' y_2' = g(t)$$

We now have a system of two linear equations for the unknowns u_1', u_2':

$$\begin{cases} u_1' y_1 + u_2' y_2 = 0 \\ u_1' y_1' + u_2' y_2' = g(t) \end{cases}$$

Using Cramer's Rule, the determinant is the Wronskian W. Thus:

$$u_1' = \frac{\det \begin{pmatrix} 0 & y_2 \\ g(t) & y_2' \end{pmatrix}}{W} = \frac{-y_2 g(t)}{W}, \qquad u_2' = \frac{\det \begin{pmatrix} y_1 & 0 \\ y_1' & g(t) \end{pmatrix}}{W} = \frac{y_1 g(t)}{W}$$

Integrating yields the result.

Example 2.5.4: Variation of Parameters
Solve $y'' + y = \sec t$.

Solution: Homogeneous solution: $y_1 = \cos t, y_2 = \sin t$. $W = 1$.

$$u_1(t) = -\int \sin t \sec t\,dt = -\int \tan t\,dt = -\ln|\sec t|$$

$$u_2(t) = \int \cos t \sec t\,dt = \int 1\,dt = t$$

Particular solution: $y_p = (-\ln|\sec t|)\cos t + (t)\sin t$.

3

Systems of Linear Equations

A scalar equation tracks one quantity, and most models track several at once: interacting species, coupled oscillators, the concentrations in a reaction. What couples them is that each rate depends on the others, so no one of the quantities satisfies an equation of its own. Collecting the unknowns into a vector $\mathbf{x}(t)$ restores a single equation, $\mathbf{x}' = A\mathbf{x}$, now with matrix-valued coefficients.

The scalar solution $y' = ay$, $y = e^{at}y_0$, then generalizes without change of form: the matrix exponential e^{At} solves the system for every A. The content of the chapter is computing e^{At}, which is where the eigenstructure of A enters, and where a defective A stops being a technicality and forces the Jordan form.

3.1 Linear Systems and Geometry

We define a first-order system of dimension n using vector notation. Let $\mathbf{x}(t) \in \mathbb{R}^n$.

Definition 3.1.1: Linear System

A first-order *linear system* is an equation of the form:

$$\mathbf{x}'(t) = A(t)\mathbf{x}(t) + \mathbf{f}(t)$$

where $A(t)$ is an $n \times n$ matrix of continuous functions, and $\mathbf{f}(t)$ is a vector of continuous forcing functions. If $\mathbf{f}(t) \equiv \mathbf{0}$, the system is *homogeneous*.

Any n-th order scalar linear ODE $y^{(n)} + p_{n-1}y^{(n-1)} + \cdots + p_0 y = 0$ can be converted into a first-order system of dimension n by introducing variables $x_1 = y, x_2 = y', \ldots, x_n = y^{(n-1)}$.

Theorem 3.1.2: Existence and Uniqueness for Systems

If $A(t)$ and $\mathbf{f}(t)$ are continuous on an interval I containing t_0, then for any initial vector $\mathbf{x}_0 \in \mathbb{R}^n$, the IVP:

$$\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}(t), \qquad \mathbf{x}(t_0) = \mathbf{x}_0$$

has a unique solution $\mathbf{x}(t)$ defined on the entire interval I.

***Proof*:**
We apply the Picard-Lindelöf Theorem for systems. We define the operator $T[\mathbf{x}](t) = \mathbf{x}_0 + \int_{t_0}^{t} (A(s)\mathbf{x}(s) + \mathbf{f}(s))\, ds$. Since the matrix entries are continuous on a closed sub-interval, they are bounded by some M. The function $F(t, \mathbf{x}) = A(t)\mathbf{x} + \mathbf{f}(t)$ is Lipschitz continuous in $\mathbf{x}$ because:

$$\|F(t, \mathbf{x}) - F(t, \mathbf{y})\| = \|A(t)(\mathbf{x} - \mathbf{y})\| \le \|A(t)\|\|\mathbf{x} - \mathbf{y}\| \le M\|\mathbf{x} - \mathbf{y}\|$$

Thus, the contraction mapping principle guarantees a unique fixed point.

Example 3.1.3: Converting Scalar to System
Convert the damped harmonic oscillator $y'' + 2y' + 2y = 0$ into a system.

Solution: Let $x_1 = y$ and $x_2 = y'$. Then $x_1' = x_2$. From the ODE, $x_2' = y'' = -2y' - 2y = -2x_2 - 2x_1$. The system is:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

3.2 The Matrix Exponential

For the scalar case $x' = ax$, the solution is $x(t) = x_0 e^{at}$. We generalize this to systems $\mathbf{x}' = A\mathbf{x}$ (where A is a constant matrix) by defining the exponential of a matrix.

Definition 3.2.1: Matrix Exponential

For an $n \times n$ matrix A, the *matrix exponential* is defined by the convergent power series:

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

This series converges absolutely for any matrix A (under any matrix norm), making e^A well-defined. Crucially, if $AB = BA$, then $e^{A+B} = e^A e^B$. However, if matrices do not commute, this identity fails.

Theorem 3.2.2: Fundamental Matrix Solution

The matrix function $\Phi(t) = e^{At}$ is the unique solution to the matrix IVP:

$$\Phi'(t) = A\Phi(t), \qquad \Phi(0) = I$$

Consequently, the solution to the vector IVP $\mathbf{x}' = A\mathbf{x}, \mathbf{x}(0) = \mathbf{x}_0$ is given by:

$$\mathbf{x}(t) = e^{At}\mathbf{x}_0$$

***Proof*:**
We differentiate the power series term by term (justified by uniform convergence):

$$\begin{aligned}
\frac{d}{dt}e^{At} &= \frac{d}{dt}\left(I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \dots\right) \\
&= 0 + A + \frac{A^2(2t)}{2!} + \frac{A^3(3t^2)}{3!} + \dots \\
&= A + A^2t + \frac{A^3t^2}{2!} + \dots \\
&= A\left(I + At + \frac{A^2t^2}{2!} + \dots\right) = Ae^{At}
\end{aligned}$$

At $t = 0$, the series becomes $I + 0 + \dots = I$. By the Uniqueness Theorem, this is the unique solution.

Example 3.2.3: Computing Matrix Exponential (Diagonal Case)

Compute e^{At} for $A = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$.

Solution: Since A is diagonal, $A^k = \begin{pmatrix} 2^k & 0 \\ 0 & (-3)^k \end{pmatrix}$. Substituting into the series:

$$e^{At} = \sum \frac{t^k}{k!}\begin{pmatrix} 2^k & 0 \\ 0 & (-3)^k \end{pmatrix} = \begin{pmatrix} \sum \frac{(2t)^k}{k!} & 0 \\ 0 & \sum \frac{(-3t)^k}{k!} \end{pmatrix} = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{-3t} \end{pmatrix}$$

3.3 Eigenvalues and Diagonalization

In practice, we do not compute infinite sums. We use eigenvalues to diagonalize the matrix.

Theorem 3.3.1: Diagonalizable Case

If A has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$, then A is diagonalizable: $A = PDP^{-1}$, where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $P = [\mathbf{v}_1 \dots \mathbf{v}_n]$.
The solution is:

$$e^{At} = Pe^{Dt}P^{-1}$$

Explicitly, the general solution is:

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t}\mathbf{v}_1 + \cdots + c_n e^{\lambda_n t}\mathbf{v}_n$$

***Proof*:**
Since $A = PDP^{-1}$, we have $A^k = (PDP^{-1})(PDP^{-1})\cdots = PD^kP^{-1}$. Thus:

$$e^{At} = \sum \frac{t^k}{k!} PD^kP^{-1} = P\left(\sum \frac{(Dt)^k}{k!}\right)P^{-1} = Pe^{Dt}P^{-1}$$

e^{Dt} is simply the diagonal matrix with entries $e^{\lambda_i t}$.

Example 3.3.2: Real Distinct Eigenvalues

Solve $\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}\mathbf{x}$.

Solution: Find eigenvalues: $\det(A - \lambda I) = (1-\lambda)^2 - 4 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$. Roots: $\lambda_1 = 3, \lambda_2 = -1$. Eigenvectors: For $\lambda_1 = 3$: $\begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix}\mathbf{v} = 0 \implies \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. For $\lambda_2 = -1$: $\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}\mathbf{v} = 0 \implies \mathbf{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$. General Solution:

$$\mathbf{x}(t) = c_1 e^{3t}\begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

3.4 Defective Matrices and Jordan Forms

What if A does not have n independent eigenvectors? This occurs when eigenvalues are repeated and the geometric multiplicity is less than the algebraic multiplicity.

We can decompose any matrix A as $A = \lambda I + N$, where N is nilpotent (meaning $N^k = 0$ for some k). Since λI and N commute, $e^{At} = e^{\lambda It}e^{Nt} = e^{\lambda t}e^{Nt}$. Because N is nilpotent, the series for e^{Nt} *truncates* to a finite polynomial!

Definition 3.4.1: Generalized Eigenvector

A vector $\mathbf{v}$ is a *generalized eigenvector* of rank k corresponding to eigenvalue λ if:

$$(A - \lambda I)^k\mathbf{v} = \mathbf{0} \quad \text{but} \quad (A - \lambda I)^{k-1}\mathbf{v} \neq \mathbf{0}$$

Theorem 3.4.2: Solutions for Defective Matrices

If an eigenvalue λ has algebraic multiplicity 2 but only 1 eigenvector $\mathbf{v}$, we find a second generalized eigenvector $\mathbf{w}$ such that:

$$(A - \lambda I)\mathbf{w} = \mathbf{v}$$

The two linearly independent solutions are:

$$\mathbf{x}_1(t) = e^{\lambda t}\mathbf{v}$$

$$\mathbf{x}_2(t) = e^{\lambda t}(\mathbf{w} + t(A - \lambda I)\mathbf{w}) = e^{\lambda t}(\mathbf{w} + t\mathbf{v})$$

***Proof*:**
We compute $e^{At}\mathbf{w}$. Since $A\mathbf{w} = \lambda\mathbf{w} + \mathbf{v}$, we recall the decomposition $A = \lambda I + (A - \lambda I)$.

$$e^{At}\mathbf{w} = e^{\lambda t}e^{(A-\lambda I)t}\mathbf{w} = e^{\lambda t}\left(I + t(A - \lambda I) + \frac{t^2}{2}(A - \lambda I)^2 + \dots\right)\mathbf{w}$$

Since $\mathbf{w}$ is a generalized eigenvector of rank 2, $(A - \lambda I)^2\mathbf{w} = (A - \lambda I)\mathbf{v} = 0$. The series terminates after the second term:

$$e^{At}\mathbf{w} = e^{\lambda t}(\mathbf{w} + t\mathbf{v})$$

This proves $\mathbf{x}_2$ is a solution.

Example 3.4.3: Repeated Eigenvalues

Solve $\mathbf{x}' = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}\mathbf{x}$.

Solution: Char. Poly: $(\lambda - 1)(\lambda - 3) + 1 = \lambda^2 - 4\lambda + 4 = (\lambda - 2)^2$. $\lambda = 2$. Eigenvector: $(A - 2I)\mathbf{v} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}\mathbf{v} = 0 \implies \mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Only one eigenvector. We need a generalized eigenvector $\mathbf{w}$:

$$(A - 2I)\mathbf{w} = \mathbf{v} \implies \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$-w_1 - w_2 = 1$. Let $w_1 = 0$, then $w_2 = -1$. $\mathbf{w} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$. General Solution:

$$\mathbf{x}(t) = c_1e^{2t}\begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2e^{2t}\left(\begin{pmatrix} 0 \\ -1 \end{pmatrix} + t\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right)$$

3.5 Phase Portraits and Stability

For 2×2 systems, we can classify the qualitative behavior of solutions based on the trace $\tau = \mathrm{tr}(A)$ and determinant $\Delta = \det(A)$. The characteristic equation is $\lambda^2 - \tau\lambda + \Delta = 0$.

- *Saddle Point ($\Delta < 0$):* Real eigenvalues with opposite signs. Unstable.
- *Node ($\Delta > 0, \tau^2 - 4\Delta > 0$):* Real eigenvalues with same sign.
  - Stable (Sink) if $\tau < 0$.
  - Unstable (Source) if $\tau > 0$.
- *Spiral ($\Delta > 0, \tau^2 - 4\Delta < 0$):* Complex eigenvalues $\alpha \pm i\beta$.
  - Stable Spiral if $\tau < 0$ ($\alpha < 0$).
  - Unstable Spiral if $\tau > 0$ ($\alpha > 0$).
  - Center if $\tau = 0$ (Purely imaginary eigenvalues).

Example 3.5.1: Classifying Stability

Classify the origin for the system with $A = \begin{pmatrix} 0 & 1 \\ -5 & -2 \end{pmatrix}$.

Solution: Trace $\tau = -2$. Determinant $\Delta = 5$. Discriminant: $\tau^2 - 4\Delta = 4 - 20 = -16 < 0$. Complex eigenvalues. Since $\tau = -2 < 0$, the real part is negative. Classification: *Stable Spiral*.

4

Qualitative Theory and Stability

Most differential equations, and nearly all non-linear ones, cannot be solved in closed form. Giving up on explicit formulae loses less than it appears to, because the questions a model is usually asked are not questions about formulae: whether the system settles to an equilibrium, oscillates, or runs away, and whether a small error in the initial state stays small. All three are questions about the shape of the solution curves.

The phase plane displays that shape directly, plotting solutions as trajectories rather than graphs. Lyapunov's definitions make stability precise without producing a single solution. And near an equilibrium, linearization replaces a non-linear system by the constant-coefficient system of chapter 3, whose behaviour is already known; the theorems of this chapter say when that replacement is legitimate.

4.1 Autonomous Systems and Phase Space

We focus on *autonomous* systems, where the governing laws do not change with time.

Definition 4.1.1: Autonomous System

A system of ODEs is *autonomous* if it can be written as:

$$\mathbf{x}' = \mathbf{f}(\mathbf{x})$$

where $\mathbf{f} : U \to \mathbb{R}^n$ is a C^1 vector field on an open set $U \subseteq \mathbb{R}^n$. The independent variable t does not appear explicitly on the right-hand side.

Definition 4.1.2: Critical Point

A point $\mathbf{x}_0$ is a *critical point* (or equilibrium point) if $\mathbf{f}(\mathbf{x}_0) = \mathbf{0}$.
If the system starts at a critical point ($\mathbf{x}(0) = \mathbf{x}_0$), the unique solution is the constant function $\mathbf{x}(t) \equiv \mathbf{x}_0$ for all t.

The set of all solution curves (trajectories) in the domain U forms the *Phase Portrait*. For autonomous systems, trajectories never cross (by Uniqueness).

Intuitively, an equilibrium is "stable" if small disturbances do not grow over time. We formalize this using $\epsilon - \delta$ language.

Definition 4.1.3: Types of Stability

Let $\mathbf{x}_0$ be a critical point.

1. *Stable (in the sense of Lyapunov):* For every $\epsilon > 0$, there exists a $\delta > 0$ such that if $\|\mathbf{x}(0) - \mathbf{x}_0\| < \delta$, then $\|\mathbf{x}(t) - \mathbf{x}_0\| < \epsilon$ for all $t \geq 0$.
2. *Asymptotically Stable:* The point is stable, and there exists a $\delta_0 > 0$ such that if $\|\mathbf{x}(0) - \mathbf{x}_0\| < \delta_0$, then $\lim_{t\to\infty} \mathbf{x}(t) = \mathbf{x}_0$.
3. *Unstable:* The point is not stable.

Stability means the solution stays close. Asymptotic stability means it stays close *and* eventually converges to the equilibrium.

For a non-linear system $\mathbf{x}' = \mathbf{f}(\mathbf{x})$, the behavior near a critical point $\mathbf{x}_0$ is often well-approximated by the linear term of its Taylor series.

$$\mathbf{f}(\mathbf{x}) \approx \mathbf{f}(\mathbf{x}_0) + D\mathbf{f}(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) = J(\mathbf{x} - \mathbf{x}_0)$$

where $J = D\mathbf{f}(\mathbf{x}_0)$ is the *Jacobian matrix* evaluated at the critical point.

Theorem 4.1.4: Principle of Linearized Stability

Let $\mathbf{x}' = \mathbf{f}(\mathbf{x})$ have a critical point at $\mathbf{0}$ (w.l.o.g.). Let $J = D\mathbf{f}(\mathbf{0})$.

1. If all eigenvalues of J have strictly negative real parts ($\mathrm{Re}(\lambda_i) < 0$), then $\mathbf{0}$ is *asymptotically stable.*
2. If at least one eigenvalue has a strictly positive real part ($\mathrm{Re}(\lambda_i) > 0$), then $\mathbf{0}$ is *unstable.*
3. If some eigenvalues have zero real part (and none positive), the linearization is inconclusive (the center case).

***Proof*:**
We sketch the proof for (1). Since all eigenvalues have negative real parts, there exists a specific inner product $\langle \cdot, \cdot \rangle_*$ and norm $\| \cdot \|_*$ such that $\langle J\mathbf{x}, \mathbf{x} \rangle_* \leq -c\|\mathbf{x}\|_*^2$ for some $c > 0$. Consider the squared norm $V(t) = \|\mathbf{x}(t)\|_*^2$. Its derivative along trajectories is:

$$\frac{d}{dt}\|\mathbf{x}\|_*^2 = 2\langle \mathbf{x}', \mathbf{x} \rangle_* = 2\langle J\mathbf{x} + \mathbf{g}(\mathbf{x}), \mathbf{x} \rangle_*$$

where $\mathbf{g}(\mathbf{x})$ represents higher-order terms ($\|\mathbf{g}(\mathbf{x})\|/\|\mathbf{x}\| \to 0$).

$$\frac{dV}{dt} \leq -2c\|\mathbf{x}\|_*^2 + 2\|\mathbf{g}(\mathbf{x})\|_*\|\mathbf{x}\|_*$$

Near $\mathbf{0}$, the quadratic decay dominated the higher-order terms, proving $V(t)$ decreases, so $\mathbf{x}(t) \to \mathbf{0}$.

Example 4.1.5: Linearization Analysis
Analyze the stability of the critical points for the pendulum equation with damping:

$$\theta'' + \gamma\theta' + \sin\theta = 0, \quad \gamma > 0$$

Solution: Convert to a system: $x = \theta, y = \theta'$.

$$\begin{cases} x' = y \\ y' = -\sin x - \gamma y \end{cases}$$

Critical points occur where $y = 0$ and $\sin x = 0$, so $x = n\pi$. The Jacobian is:

$$J(x, y) = \begin{pmatrix} 0 & 1 \\ -\cos x & -\gamma \end{pmatrix}$$

At $(0, 0)$ (downward position): $J = \begin{pmatrix} 0 & 1 \\ -1 & -\gamma \end{pmatrix}$. Char. Poly: $\lambda^2 + \gamma\lambda + 1 = 0$. Roots $\lambda = \frac{-\gamma \pm \sqrt{\gamma^2 - 4}}{2}$. Real parts are negative. *Asymptotically Stable.*

At $(\pi, 0)$ (upward position): $J = \begin{pmatrix} 0 & 1 \\ 1 & -\gamma \end{pmatrix}$. Char. Poly: $\lambda^2 + \gamma\lambda - 1 = 0$. Roots $\lambda = \frac{-\gamma \pm \sqrt{\gamma^2 + 4}}{2}$. One root is positive. *Unstable Saddle.*

4.2 Lyapunov's Direct Method

Linearization fails when eigenvalues have zero real parts. Lyapunov's Direct Method (or Second Method) allows us to determine stability without solving the equation, by generalizing the concept of energy.

Definition 4.2.1: Lyapunov Function

Let $\mathbf{x}^*$ be a critical point. A scalar function $V : U \to \mathbb{R}$ is a *Lyapunov Function* if:

1. $V(\mathbf{x}^*) = 0$ and $V(\mathbf{x}) > 0$ for $\mathbf{x} \neq \mathbf{x}^*$ (Positive Definite).
2. $\dot{V}(\mathbf{x}) = \nabla V \cdot \mathbf{f}(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in U \setminus \{\mathbf{x}^*\}$ (Orbital Derivative is Negative Semi-Definite).

If the inequality in (2) is strict ($\dot{V} < 0$), it is a *Strict Lyapunov Function.*

Theorem 4.2.2: Lyapunov Stability Theorem

Let $\mathbf{x}^*$ be a critical point.

1. If there exists a Lyapunov Function V in a neighborhood of $\mathbf{x}^*$, then $\mathbf{x}^*$ is *Stable*.
2. If there exists a Strict Lyapunov Function V, then $\mathbf{x}^*$ is *Asymptotically Stable*.

***Proof*:**
We prove (1). Let $\epsilon > 0$ be small enough that the ball $B_\epsilon(\mathbf{x}^*)$ is in the domain of V. Let $m = \min_{\|\mathbf{x}\|=\epsilon} V(\mathbf{x})$. Since V is continuous and positive definite, $m > 0$. Define the set $U = \{\mathbf{x} \big| V(\mathbf{x}) < m\}$. Since $V(\mathbf{x}^*) = 0$ and V is continuous, there exists a $\delta > 0$ such that $B_\delta(\mathbf{x}^*) \subset U$.

Now, let a trajectory start at $\mathbf{x}_0$ with $\|\mathbf{x}_0 - \mathbf{x}^*\| < \delta$. Then $V(\mathbf{x}_0) < m$. Since $\dot{V} \leq 0$, $V(\mathbf{x}(t))$ is non-increasing. Thus, $V(\mathbf{x}(t)) \leq V(\mathbf{x}_0) < m$ for all $t \geq 0$. Since the boundary of B_ϵ has $V \geq m$, the trajectory can never reach the boundary $\|\mathbf{x}\| = \epsilon$. Therefore, $\|\mathbf{x}(t) - \mathbf{x}^*\| < \epsilon$ for all t. This is the definition of stability.

Example 4.2.3: Lyapunov Function
Show that the origin is stable for the system:

$$x' = -x - xy^2, \qquad y' = -y - yx^2$$

Solution: Try $V(x, y) = x^2 + y^2$. This is positive definite. Compute $\dot{V}$:

$$\begin{aligned}
\dot{V} &= \frac{\partial V}{\partial x}x' + \frac{\partial V}{\partial y}y' \\
&= 2x(-x - xy^2) + 2y(-y - yx^2) \\
&= -2x^2 - 2x^2y^2 - 2y^2 - 2y^2x^2 \\
&= -2(x^2 + y^2) - 4x^2y^2
\end{aligned}$$

Clearly $\dot{V} < 0$ for all $(x, y) \neq (0, 0)$. Thus, V is a strict Lyapunov function, and the origin is asymptotically stable.

4.3 Limit Cycles

In non-linear systems, solutions can approach a closed periodic orbit called a *limit cycle*.

Theorem 4.3.1: Poincaré-Bendixson Theorem

Suppose R is a closed, bounded region in the plane ($\mathbb{R}^2$) that contains no critical points. If a trajectory $\mathbf{x}(t)$ is confined in R for all $t \geq 0$, then $\mathbf{x}(t)$ must either be a periodic orbit or approach a periodic orbit as $t \to \infty$.

***Proof*:**
(Sketch) The proof relies on the Jordan Curve Theorem (unique to $\mathbb{R}^2$). Since trajectories cannot cross, a bounded trajectory that doesn't terminate at a fixed point must eventually wind onto itself, forming a limit cycle.

Example 4.3.2: Existence of Limit Cycle
Consider $r' = r(1 - r^2), \theta' = 1$ (in polar coordinates).

- If $r < 1$, $r' > 0$ (radius increases).
- If $r > 1$, $r' < 0$ (radius decreases).

Consider the annulus $R = \{1/2 \leq r \leq 2\}$. Trajectories enter R on both boundaries and never leave. Since there are no critical points (as $\theta' = 1 \neq 0$), any solution starting in R must approach the limit cycle at $r = 1$.

5

Boundary Value Problems and Sturm-Liouville Theory

Every problem so far has specified its conditions at a single point t_0. Imposing them instead at the two ends of an interval changes the theory rather than the technique. A linear initial value problem always has exactly one solution; a boundary value problem may have none, exactly one, or infinitely many, and which of the three occurs depends on the parameter in the equation only through the spectrum of the differential operator.

Solvability therefore becomes an eigenvalue question, and Sturm-Liouville theory is the eigenvalue theory that answers it. It is diagonalization transplanted from matrices to differential operators: a self-adjoint operator, real eigenvalues, and an orthogonal basis of eigenfunctions in which a solution may be expanded. Green's functions complete the picture by inverting the operator outright on the inhomogeneous problem.

5.1 Introduction to Boundary Value Problems

Consider the second-order linear equation on an interval $[a, b]$:

$$L[y] = y'' + p(x)y' + q(x)y = g(x), \qquad a < x < b$$

subject to boundary conditions at the endpoints:

$$B_1[y] = \alpha_1 y(a) + \alpha_2 y'(a) = \gamma_1, \qquad B_2[y] = \beta_1 y(b) + \beta_2 y'(b) = \gamma_2$$

Definition 5.1.1: Homogeneous BVP

A BVP is called *homogeneous* if both the differential equation ($g(x) \equiv 0$) and the boundary conditions ($\gamma_1 = \gamma_2 = 0$) are homogeneous.

The trivial solution $y(x) \equiv 0$ is always a solution to a homogeneous BVP. The central question of this chapter is: *Under what conditions does a non-trivial solution exist?*

Example 5.1.2: Eigenvalues of a BVP

Consider the problem:

$$y'' + \lambda y = 0, \qquad y(0) = 0, \quad y(\pi) = 0$$

Depending on the parameter λ, the nature of the solution changes:

1. If $\lambda < 0$ (let $\lambda = -\mu^2$): $y(x) = c_1 \cosh \mu x + c_2 \sinh \mu x$. The boundary conditions force $c_1 = c_2 = 0$.
2. If $\lambda = 0$: $y(x) = c_1 x + c_2$. Boundary conditions force $c_1 = c_2 = 0$.
3. If $\lambda > 0$ (let $\lambda = \mu^2$): $y(x) = c_1 \cos \mu x + c_2 \sin \mu x$. $y(0) = 0 \implies c_1 = 0$. $y(\pi) = c_2 \sin(\mu\pi) = 0$. For a non-trivial solution, we require $\sin(\mu\pi) = 0$, which implies $\mu = n$ for $n \in \mathbb{Z}^+$.

Thus, non-trivial solutions (eigenfunctions) $y_n(x) = \sin(nx)$ exist only for the discrete spectrum $\lambda_n = n^2$.

To understand the existence of eigenvalues generally, we must define the "transpose" of a differential operator.

Definition 5.1.3: Formal Adjoint

For the operator $L[y] = p_2(x)y'' + p_1(x)y' + p_0(x)y$, the *formal adjoint* L^* is defined by:

$$L^*[u] = \frac{d^2}{dx^2}(p_2 u) - \frac{d}{dx}(p_1 u) + p_0 u$$

The operator L is *formally self-adjoint* if $L = L^*$.

A straightforward calculation shows that L is self-adjoint if and only if $p_1(x) = p_2'(x)$. This motivates the *Sturm-Liouville form.*

Definition 5.1.4: Sturm-Liouville Form

A second-order differential operator L is in *Sturm-Liouville form* if:

$$L[y] = -\frac{d}{dx}\left[p(x)\frac{dy}{dx}\right] + q(x)y$$

where $p(x) > 0$. This operator is formally self-adjoint.

Note: Any linear operator $P(x)y'' + Q(x)y' + R(x)y$ can be multiplied by the integrating factor

$\mu(x) = \frac{1}{P(x)} e^{\int Q/P dx}$ to be converted into Sturm-Liouville form.

Theorem 5.1.5: Lagrange's Identity

For a self-adjoint operator L in Sturm-Liouville form:

$$uL[v] - vL[u] = \frac{d}{dx}\left[p(x)(uv' - u'v)\right]$$

Integrating over $[a, b]$ yields Green's Formula:

$$\langle L[u], v\rangle - \langle u, L[v]\rangle = \left[p(x)(uv' - u'v)\right]_a^b$$

If u and v satisfy *separated boundary conditions* (e.g., $\alpha_1 u(a) + \alpha_2 u'(a) = 0$), the boundary term vanishes, and we obtain the crucial symmetry property: $\langle L[u], v\rangle = \langle u, L[v]\rangle$.

5.2 The Sturm-Liouville Theorems

We now consider the eigenvalue problem:

$$L[y] = \lambda w(x) y$$

subject to homogeneous separated boundary conditions.

Theorem 5.2.1: Sturm-Liouville Theorem

For a regular Sturm-Liouville problem:

1. All eigenvalues λ are real.
2. Eigenfunctions φ_n, φ_m corresponding to distinct eigenvalues are *orthogonal* with respect to the weight $w(x)$:

$$\int_a^b \varphi_n(x)\varphi_m(x)w(x)\, dx = 0 \quad \text{if } n \neq m$$

3. The eigenvalues form an infinite sequence $\lambda_1 < \lambda_2 < \dots$ with $\lim_{n\to\infty} \lambda_n = \infty$.
4. The eigenfunctions form a *complete basis* for the space of piecewise continuous functions $L^2_w[a, b]$.

***Proof*:**
Real Eigenvalues: Let $L[u] = \lambda w u$. Take the complex conjugate $L[\bar{u}] = \bar{\lambda} w \bar{u}$. Using Green's Formula (with boundary terms vanishing):

$$(\lambda - \bar{\lambda})\langle wu, \bar{u}\rangle = \langle L[u], \bar{u}\rangle - \langle u, L[\bar{u}]\rangle = 0$$

Since $w(x) > 0$, $\langle wu, \bar{u}\rangle = \int w|u|^2 > 0$, implying $\lambda = \bar{\lambda}$.

Orthogonality: Let $L[\varphi_n] = \lambda_n w \varphi_n$ and $L[\varphi_m] = \lambda_m w \varphi_m$.

$$(\lambda_n - \lambda_m)\langle w\varphi_n, \varphi_m\rangle = \langle L[\varphi_n], \varphi_m\rangle - \langle \varphi_n, L[\varphi_m]\rangle = 0$$

If $\lambda_n \neq \lambda_m$, the inner product must vanish.

Theorem 5.2.2: Eigenfunction Expansion

Any piecewise smooth function $f(x)$ on $[a, b]$ can be represented as:

$$f(x) \sim \sum_{n=1}^{\infty} c_n \varphi_n(x)$$

where the Fourier coefficients are given by the projection:

$$c_n = \frac{\langle f, \varphi_n\rangle_w}{\langle \varphi_n, \varphi_n\rangle_w} = \frac{\int_a^b f(x)\varphi_n(x)w(x)\,dx}{\int_a^b \varphi_n(x)^2 w(x)\,dx}$$

5.3 The Fredholm Alternative

We now address the inhomogeneous equation $L[y] - \lambda w(x)y = f(x)$. The solvability depends on whether λ is an eigenvalue.

Theorem 5.3.1: Fredholm Alternative

Consider the inhomogeneous BVP $L[y] - \sigma w(x)y = f(x)$.

1. **Non-Resonant Case:** If σ is *not* an eigenvalue of the homogeneous problem, there exists a unique solution for any $f(x)$.

2. **Resonant Case:** If $\sigma = \lambda_k$ is an eigenvalue with eigenfunction φ_k, a solution exists *if and only if* f is orthogonal to the eigenfunction:

$$\int_a^b f(x)\varphi_k(x)w(x)\,dx = 0$$

If this condition holds, there are infinitely many solutions (of the form $y_p + c\varphi_k$).

5.4 Green's Functions

In the non-resonant case, the unique solution can be represented by an integral kernel called the Green's function.

Definition 5.4.1: Green's Function

The *Green's Function* $G(x, \xi)$ for the operator L is the solution to the distributional equation:

$$L[G(x, \xi)] = \delta(x - \xi)$$

subject to the homogeneous boundary conditions.

To construct $G(x, \xi)$ explicitly, we solve the homogeneous equation $L[y] = 0$ on the intervals $[a, \xi)$ and $(\xi, b]$. Let $y_1(x)$ satisfy the boundary condition at a, and $y_2(x)$ satisfy the condition at b. We assume:

$$G(x, \xi) = \begin{cases} c_1(\xi) y_1(x) & a \le x < \xi \\ c_2(\xi) y_2(x) & \xi < x \le b \end{cases}$$

We require two conditions at $x = \xi$:

1. *Continuity:* $c_1 y_1(\xi) = c_2 y_2(\xi)$.
2. *Jump in Derivative:* Integrating $-\frac{d}{dx}[py'] = \delta$ from $\xi - \epsilon$ to $\xi + \epsilon$ implies a jump of $-1/p(\xi)$:

$$\frac{\partial G}{\partial x}(\xi^+, \xi) - \frac{\partial G}{\partial x}(\xi^-, \xi) = -\frac{1}{p(\xi)}$$

Solving this system for c_1, c_2 using the Wronskian $W(\xi) = y_1 y_2' - y_2 y_1'$ yields the symmetric form:

Theorem 5.4.2: Structure of the Green's Function

The Green's function is given by:

$$G(x, \xi) = \frac{1}{p(\xi) W(\xi)} \begin{cases} y_1(x) y_2(\xi) & a \le x \le \xi \\ y_1(\xi) y_2(x) & \xi \le x \le b \end{cases}$$

Note that $p(\xi) W(\xi)$ is a constant. The solution to $L[y] = f(x)$ is then:

$$y(x) = \int_a^b G(x, \xi) f(\xi) \, d\xi$$

6

Series Solutions and Special Functions

The linear equations solved so far have had constant coefficients, which is what allowed exponentials to reduce them to polynomials. The equations of mathematical physics do not cooperate. Separating the heat, wave, and Schrödinger equations in cylindrical or spherical coordinates produces ordinary equations whose coefficients vary with the independent variable, and these almost never have solutions expressible in e^x, $\sin x$, and the rest of the elementary vocabulary.

The response is to widen what counts as a solution. A power series determined term by term from the equation defines a function whether or not that function has an elementary name, and the Frobenius method extends the construction across the singular points where the leading coefficient vanishes, which is exactly where the physical problems place them. The functions this produces, Bessel and Legendre chief among them, are the standard vocabulary for those problems.

6.1 Analytic Functions and Ordinary Points

We consider the homogeneous linear second-order equation:

$$P(x)y'' + Q(x)y' + R(x)y = 0 \tag{6.1}$$

Dividing by $P(x)$, we obtain the standard form $y'' + p(x)y' + q(x)y = 0$. The behavior of solutions depends entirely on the behavior of $p(x)$ and $q(x)$ at a given point x_0.

Definition 6.1.1: Ordinary and Singular Points

Let $p(x) = Q(x)/P(x)$ and $q(x) = R(x)/P(x)$. A point x_0 is called an *ordinary point* if both $p(x)$ and $q(x)$ are analytic at x_0 (i.e., they have convergent Taylor series expansions in a neighborhood of x_0).

If either function is not analytic at x_0, the point is called a *singular point*.

Since polynomials are analytic everywhere, if P, Q, R are polynomials, singular points only occur at the roots of $P(x) = 0$.

Theorem 6.1.2: Existence of Power Series Solutions

If x_0 is an ordinary point of the differential equation, then there exist two linearly independent solutions of the form:

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

The radius of convergence of this series is at least as large as the distance from x_0 to the nearest singular point (in the complex plane).

***Proof*:**
(Sketch) Since $p(x)$ and $q(x)$ are analytic, they can be represented by power series. We assume $y(x) = \sum a_n (x - x_0)^n$, differentiate term-by-term, and substitute into the ODE. Equating coefficients of like powers of $(x - x_0)$ yields a recurrence relation determining a_n for $n \geq 2$ in terms of the arbitrary constants a_0, a_1. The convergence is guaranteed by the theory of analytic functions in complex analysis.

Example 6.1.3: Legendre's Equation

Consider Legendre's equation, which arises in spherical coordinates:

$$(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0$$

Here $P(x) = 1 - x^2$. The singular points are $x = \pm 1$. The point $x_0 = 0$ is an ordinary point.

Solution: Let $y = \sum a_n x^n$.

$$y' = \sum n a_n x^{n-1}, \qquad y'' = \sum n(n-1) a_n x^{n-2}$$

Substituting into the ODE:

$$(1 - x^2) \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - 2x \sum_{n=1}^{\infty} n a_n x^{n-1} + \alpha(\alpha + 1) \sum_{n=0}^{\infty} a_n x^n = 0$$

Shifting indices to align powers of x^k:

$$\sum_{k=0}^{\infty} \left[(k+2)(k+1) a_{k+2} - (k(k-1) + 2k - \alpha(\alpha+1)) a_k \right] x^k = 0$$

The recurrence relation is:

$$a_{k+2} = -\frac{(\alpha - k)(\alpha + k + 1)}{(k+2)(k+1)} a_k$$

Notice that if $\alpha = n$ is a non-negative integer, the numerator vanishes when $k = n$, causing the series to terminate. These terminating solutions are the *Legendre Polynomials* $P_n(x)$.

6.2 Regular Singular Points and the Method of Frobenius

Many physical potentials (like the Coulomb potential $1/r$) introduce singularities at the origin. Standard power series fail here. We need a more robust method.

Definition 6.2.1: Regular Singular Point

A singular point x_0 is called a *regular singular point* if the singularities are no worse than a pole of order 1 for $p(x)$ and order 2 for $q(x)$. Specifically, if:

$$\lim_{x \to x_0} (x - x_0) p(x) \quad \text{and} \quad \lim_{x \to x_0} (x - x_0)^2 q(x)$$

are both finite (analytic), then x_0 is regular.

For such points, we assume a solution of the form:

$$y(x) = (x - x_0)^r \sum_{n=0}^{\infty} a_n (x - x_0)^n = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+r}, \quad a_0 \neq 0$$

This is called the *Frobenius Method*. The exponent r is determined by the "Indicial Equation."

Theorem 6.2.2: Frobenius Theorem

Let $x = 0$ be a regular singular point. Let $p_0 = \lim_{x \to 0} x p(x)$ and $q_0 = \lim_{x \to 0} x^2 q(x)$. The *Indicial Equation* is:

$$r(r-1) + p_0 r + q_0 = 0$$

Let r_1, r_2 be the roots with $\mathrm{Re}(r_1) \geq \mathrm{Re}(r_2)$.

1. There always exists one solution $y_1(x) = |x|^{r_1} \sum a_n x^n$.
2. If $r_1 - r_2$ is not an integer, the second solution is $y_2(x) = |x|^{r_2} \sum b_n x^n$.
3. If $r_1 - r_2$ is an integer (or zero), the second solution may involve a logarithm:

$$y_2(x) = C y_1(x) \ln |x| + |x|^{r_2} \sum b_n x^n$$

The most important application of the Frobenius method is solving Bessel's equation, which arises when separating variables in cylindrical coordinates (Laplacian in 2D or 3D).

Example 6.2.3: Bessel's Equation of Order ν

$$x^2y'' + xy' + (x^2 - \nu^2)y = 0, \qquad \nu \geq 0$$

Here $p(x) = 1/x$ and $q(x) = (x^2 - \nu^2)/x^2$. $x = 0$ is a regular singular point.

Indicial Equation: Multiply by x^2 to identify the leading terms: $x^2y'' + xy' - \nu^2 y \approx 0$ near $x = 0$. This is Cauchy-Euler! The characteristic equation is $r(r-1) + r - \nu^2 = 0 \implies r^2 - \nu^2 = 0 \implies r = \pm\nu$.

Solution Construction: Using $r_1 = \nu$, the Frobenius method yields the *Bessel Function of the First Kind*, denoted $J_\nu(x)$:

$$J_\nu(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\Gamma(n+\nu+1)} \left(\frac{x}{2}\right)^{2n+\nu}$$

For the second solution, since $r_1 - r_2 = 2\nu$ might be an integer, we encounter the logarithmic case. The second linearly independent solution is the *Bessel Function of the Second Kind*, $Y_\nu(x)$, which blows up at $x = 0$.

Sturm-Liouville Properties of Special Functions

Why do we care about these complicated series? Because they form orthogonal bases, just like sines and cosines.

Recall Bessel's equation: $(xy')' + (x - \frac{\nu^2}{x})y = 0$. This is almost Sturm-Liouville form with $p(x) = x, q(x) = -\nu^2/x, \lambda = 1, w(x) = x$. Consider the BVP on $[0, R]$ with $y(R) = 0$. The solutions are $J_\nu(\lambda_k x)$ where λ_k are roots of $J_\nu(\lambda R) = 0$.

Theorem 6.2.4: Orthogonality of Bessel Functions

Let $\{\lambda_k\}$ be the positive zeros of $J_\nu(x)$. Then the functions $\varphi_k(x) = J_\nu(\lambda_k x/R)$ are orthogonal on $[0, R]$ with respect to the weight $w(x) = x$:

$$\int_0^R J_\nu\left(\frac{\lambda_k x}{R}\right) J_\nu\left(\frac{\lambda_m x}{R}\right) x\,dx = 0 \qquad \text{if } k \neq m$$

This orthogonality allows us to expand any function $f(r)$ on a disk into a Fourier-Bessel series:

$$f(r) = \sum_{k=1}^{\infty} c_k J_\nu\left(\frac{\lambda_k r}{R}\right)$$

This is the fundamental mechanism for solving heat flow on a circular plate or vibrations of a drum head.

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